

Managing Processes in Linux

1. What is a Process?

A process is simply an instance of a running program. Every time you execute a command or open an application in Linux, a new process is created.

- **Example:** Opening "LibreOffice Writer" starts a process.
- **Note:** You can have multiple processes (instances) of the same program running at the same time.

2. Types of Processes

There are two main categories of processes in Linux:

A. Foreground Processes

- **Definition:** These run on the screen and require user input to function.
- **Behavior:** If you run a foreground process in the Terminal, you cannot use that Terminal window for other commands until the process finishes.
- **Example:** Office programs, Web browsers.

B. Background Processes

- **Definition:** These run behind the scenes and usually do not need user input.
- **Behavior:** They do not block the Terminal; you can continue typing commands while they run.
- **Example:** Antivirus software scanning files.

3. Managing Foreground & Background Processes

If you launch a program (like a media player) from the Terminal, it locks the terminal. Here is how to manage that:

How to send a running process to the Background:

- **Start the program:** Type the command (e.g., `banshee`).
- **Stop/Pause the process:** Press `Ctrl + Z`. This suspends the program.
- **Send to Background:** Type `bg` and hit Enter.

- **Result:** The program continues running, but your Terminal is now free for you to use again.
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- **How to bring a process back to the Foreground:**
 - **Command:** `fg [jobname]`
 - **Example:** `fg banshee`
- **Result:** Brings the background process back to the active terminal window.

4. System Monitoring Utilities

A. The “top” command

This is a real-time system monitor. It shows all running processes and updates dynamically.

- **Syntax:** `top`
- **Key Columns in Output:**
 - **PID:** Process ID (Unique number for every task).
 - **USER:** The owner of the process (e.g., root, guru99).
 - **PR:** Priority.
 - **NI:** Nice Value (explained in section 6).
 - **VIRT/RES:** Memory usage (Virtual and Physical).
 - **%CPU / %MEM:** Percentage of CPU and RAM being used.
 - **S (Status):** The state of the process.
 - **D:** Uninterruptible sleep.
 - **R:** Running.
 - **S:** Sleeping.
 - **T:** Traced or Stopped.
 - **Z:** Zombie (A process that has completed but still has an entry in the process table).

B. The “ps” command

Unlike `top`, which updates in real-time, `ps` takes a static **snapshot** of current processes (similar to Windows Task Manager).

- **Command:** `ps ux`

- **Result:** Shows all processes running for the current user with details (CPU, MEM, Start time).
- **Command:** `ps [PID]`
 - **Result:** Shows the status of one specific process ID.

C. The “pidof” command

If you know the program name but not the ID, use this.

- **Command:** `pidof [program_name]`
- **Example:** `pidof firefox`
- **Result:** Returns the number (PID), e.g., 3691.

5. Terminating Processes (The `kill` Command)

If a program is unresponsive or you need to close it from the terminal.

- **Syntax:** `kill [PID]`
- **Steps:**
 1. Find the PID: `pidof firefox` (Result: 3691)
 2. Kill it: `kill 3691`
- **Importance:** Essential for system administrators to remove stuck or malicious processes without rebooting.

6. Prioritizing Processes (Niceness)

Linux can run many processes at once. To decide which process gets more CPU power, Linux uses an index called "**Niceness**" (Nice value).

- **Range:** -20 to 19.
- **Default Value:** 0.
- **Important Rule:** The **LOWER** the number, the **HIGHER** the priority.
 - -20 = Highest Priority.
 - 19 = Lowest Priority.

A. Starting a process with a specific priority (nice)

- **Syntax:** `nice -n [value] [process_name]`

- **Example:** `nice -n 19 banshee`
- **Result:** Starts the Banshee player with the lowest priority.

B. Changing priority of a running process (renice)

- **Syntax:** `renice [value] -p [PID]`
- **Example:** `sudo renice -20 -p 3844`
- **Result:** Changes the priority of the running process (PID 3844) to the highest possible priority (-20).

Summary Table

Command	Description
<code>Ctrl + Z</code>	Stop/Suspend a foreground process.
<code>bg</code>	Move a stopped process to the Background.
<code>fg</code>	Move a process to the Foreground.
<code>top</code>	View real-time active processes.
<code>ps</code>	View a snapshot of current processes.
<code>pidof</code>	Find the ID of a running program.
<code>kill</code>	Terminate a process using its PID.

`nice`

Start a program with a specific priority.

`renice`

Change the priority of an existing process.